

where $\widehat{C}(g, t)$ is as defined in Algorithm 1.

Remark 10. In DiD applications, it is common to use “cluster-robust” inference procedures; see, e.g., Wooldridge (2003) and Bertrand et al. (2004). However, we note that the choice of whether to cluster or not is usually not obvious, and depends on the kind of uncertainty one is trying to reflect; see, e.g., Abadie et al. (2017) for a discussion in a cross-sectional setup.¹⁵ In the case that one wishes to account for clustering to reflect “cluster-based” sampling uncertainty, we note that this can be done in a straightforward manner using a small modification of the multiplier bootstrap described above, provided that the number of cluster is “large.” More precisely, instead of drawing observation-specific V ’s, one simply needs to draw cluster-specific V ’s; see, e.g., Sherman and Le Cessie (2007), Kline and Santos (2012), Cheng et al. (2013), and MacKinnon and Webb (2018, 2020). If the number of clusters is “small,” however, the application of the aforementioned bootstrap procedure is not warranted.¹⁶

Remark 11. In Algorithm 1 we have required an estimator for the main diagonal of Σ . However, we note that if one takes $\widehat{\Sigma}(g, t) = 1$ for all (g, t) , the result in Corollary 1 continues to hold. However, the resulting “constant width” simultaneous confidence band may be of larger length; see, e.g., Montiel Olea and Plagborg-Møller (2018) and Freyberger and Rai (2018).

Remark 12. The above results focus on making inference about $ATT(g, t)$ ’s in (effective) post-treatment periods $t \geq g - \delta$. Although the limited anticipation condition in Assumption 3 implies that $ATT(g, t) = 0$ for all $t < g - \delta$ regardless of the group g , it is common practice to also estimate these pre-treatment parameters and use them to assess the credibility of the underlying identifying assumptions. Note that our DiD estimands (2.2) - (2.7) can be easily adjusted to include these by simply replacing the “long differences” $(Y_t - Y_{g-\delta-1})$ with the “short differences” $(Y_t - Y_{t-1})$ for all $t < g - \delta$. All our results continues to hold when one augments $\widehat{ATT}_{t \geq (g-\delta)}^{dr, nev}$ to also include these estimates for the $ATT(g, t)$ ’s in the pre-treatment periods $t < g - \delta$.

4.2 Asymptotic Theory for Summary Parameters

Assume, for simplicity, that Assumption 3 holds with $\delta = 0$. In this section, we discuss how one can estimate and make inference about the summary measures of the casual effects discussed in Section 3. More concisely, we consider parameters of the form of θ as defined in (3.1), which covers all of the aggregated parameters discussed in Section 3.

Given the discussion in Section 4.1, a natural way to estimate θ is to use the plug-in type estimators

$$\hat{\theta} = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} \widehat{w}(g, t) \widehat{ATT}_{dr}^{nev}(g, t; 0),$$

¹⁵The formal results in Abadie et al. (2017) focus on the cross section case and rely on additional functional form restrictions that we do not impose in this paper. Fully extending the results of Abadie et al. (2017) to the semiparametric panel data case is beyond the scope of our paper.

¹⁶In such cases, provided that one is comfortable imposing additional functional form assumptions, one could use alternative procedures such as Conley and Taber (2011) and Ferman and Pinto (2019). Extending these proposals to our setup is beyond the scope of this paper though.

where $\widehat{w}(g, t)$ are estimators for $w(g, t)$ such that for all $g \in \mathcal{G}$ and $t = 2, \dots, \mathcal{T}$,

$$\sqrt{n}(\widehat{w}(g, t) - w(g, t)) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_{g,t}^w(W_i) + o_p(1),$$

with $\mathbb{E}[\xi_{gt}^w(W)] = 0$ and $\mathbb{E}[\xi_{gt}^w(W)\xi_{gt}^w(W)']$ finite and positive definite. Estimators based on the sample analogue of the weights discussed in Section 3 satisfy this condition.

Let

$$l^w(W_i) = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} \left(w(g, t) \cdot \psi_{g,t,0}^{dr,nev}(W_i; \kappa_{g,t}^{*,nev}) + \xi_{g,t}^w(W_i) \cdot ATT(g, t) \right),$$

where $\psi_{g,t,\delta}^{dr,nev}$ are as defined in (4.4).

The following result follows immediately from Theorem 2, and can be used to conduct asymptotically valid inference for the summary causal parameters θ .

Corollary 2. *Under the assumptions of Theorem 2,*

$$\begin{aligned} \sqrt{n}(\hat{\theta} - \theta) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n l^w(W_i) + o_p(1) \\ &\xrightarrow{d} N\left(0, \mathbb{E}\left[l^w(W)^2\right]\right) \end{aligned}$$

Corollary 2 implies that one can construct standard errors and confidence intervals for summary treatment effect parameters based on a consistent estimator of $\mathbb{E}[l^w(W)^2]$ or by using a bootstrap procedure like the one in Algorithm 1. The main advantage of using the bootstrap procedure akin to Algorithm 1 is that inference procedures would be robust against multiple-testing problems. This is particularly attractive when considering $\theta_{es}(e)$, $\theta_{es}^{bal}(e; e')$, $\theta_{sel}(\tilde{g})$, and $\theta_c(\tilde{t})$, as practitioners would probably analyze how these parameters differ across event-times e , groups $\tilde{g}$, and calendar-time $\tilde{t}$.

Remark 13. *As discussed in Remark 10, the validity of the “cluster-robust” multiplier bootstrap procedure relies on the number of clusters being “large.” In some applications such a condition may be more plausible when analyzing the aggregated parameter θ than when analyzing the $ATT(g, t)$ themselves.*

5 The Effect of Minimum Wage Policy on Teen Employment

In this section, we illustrate the empirical relevance of our proposed methods. To do this, we apply our methods to study the effect of the minimum wage on teen employment. The main goal of this section is to compare results arising from using a TWFE specification (as is most common in applications) to results coming from our proposed method. We think that this comparison is important in order to get a sense of whether the theoretical limitations of TWFE discussed in recent work end up translating into meaningful differences in applications. Moreover, one might expect that understanding the effect of a minimum wage change on employment is a challenging case for TWFE as the effect of the minimum wage may be dynamic (Meer and West (2016)) and the timing of minimum wage changes varies across states. Unlike TWFE, the approach that we have proposed in the current paper is robust to these challenges.

By far the most common approach to trying to understand the effect of the minimum wage on employment is to exploit variation in the timing of minimum wage increases across states. Our identification strategy follows this approach. In particular, we consider a time period from 2001-2007 where the federal minimum wage was flat at \$5.15 per hour. We focus on county level teen employment in states whose minimum wage was equal to the federal minimum wage at the beginning of the period. Some of these states increased their minimum wage over this period – these become treated groups. In particular, we define groups by the time period when a state first increased its minimum wage. Others did not increase their minimum wage – these are the untreated group. This setup allows us to have more data than local case study approaches. On the other hand, it also allows us to have cleaner identification (state-level minimum wage policy changes) than in studies with more periods; the latter setup is more complicated than ours particularly because of the variation in the federal minimum wage over time. It also allows us to check for internal consistency of identifying assumptions – namely whether or not the identifying assumptions hold in periods before particular states raised their minimum wages.

We use county level data on teen employment and other county characteristics. County level teen employment comes from the Quarterly Workforce Indicators (QWI), as in [Dube et al. \(2016\)](#); see [Dube et al. \(2016\)](#) for a detailed discussion of this dataset. Other pre-treatment county characteristics come from the 2000 County Data Book. These include county population in 2000, the fraction of the population that is white, educational characteristics from 1990, median income in 1997, and the fraction of the population below the poverty level in 1997. After dropping ten states due to their minimum wage being higher than the federal minimum wage in 2000, seven other states for lack of data on teen employment, and four other states in the Northern census region, our final sample includes county-level data from 29 states. We provide additional details on constructing the data in the Supplementary Appendix.

Summary statistics for county characteristics are provided in Table 2. There are some notable differences in county characteristics between counties in states that increased their minimum wage and in states that did not increase their minimum wage. Treated counties are much less likely to be in the South. They also have much higher population (on average 94,000 compared to 53,000 for untreated counties). The proportion of white residents is higher in treated counties (on average, 89% compared to 83% for untreated counties). There are smaller differences in the fraction with high school degrees and the poverty rate though the differences are both statistically significant. Treated counties have a somewhat higher fraction of high school graduates and a somewhat lower poverty rate.

5.1 Results

In the following we discuss different sets of results using different identification strategies. In particular, we consider the cases in which one would assume that the parallel trends assumption would hold unconditionally, and when it holds only after controlling on observed characteristics X . In the main text, we consider the case where never-treated counties are the comparison group and where we do not allow for any anticipation effects (i.e., $\delta = 0$). We provide results using the not-yet-treated counties as the comparison group and allowing for one year anticipation in the Supplementary Appendix; results from those cases are quite similar to the ones presented here.

The first set of results comes from using the unconditional parallel trends assumption to estimate the

Table 2: Summary Statistics for Main Dataset

	Treated Counties	Untreated Counties	Diff.	P-val on Diff.
Midwest	0.59	0.34	0.25	0.00
South	0.27	0.59	-0.32	0.00
West	0.14	0.07	0.07	0.00
Population (1000s)	94.32	53.43	40.89	0.00
White	0.89	0.83	0.06	0.00
HS Graduates	0.59	0.55	0.04	0.00
Poverty Rate	0.13	0.16	-0.03	0.00
Median Inc. (1000s)	33.91	31.89	2.02	0.00

Notes: Summary statistics for counties located in states that raised their minimum wage between Q2 of 2003 and Q1 of 2007 (treated) and states whose minimum wage was effectively set at the federal minimum wage for the entire period (untreated). The sample consists of 2284 counties. *Sources:* Quarterly Workforce Indicators and 2000 County Data Book

effect of raising the minimum wage on teen employment. The results for group-time average treatment effects are reported in Panel (a) of Figure 1 along with a uniform 95% confidence band. All inference procedures use clustered bootstrapped standard errors at the county level, and account for the autocorrelation of the data. The plot contains pre-treatment estimates that can be used to “pre-test” the parallel trends assumption as well as treatment effect estimates in post-treatment periods.

The group-time average treatment effect estimates provide support for the view that increasing the minimum wage led to a reduction in teen employment. For 5 out of 7 group-time average treatment effects, there is a clear statistically significant negative effect on employment. The other two are marginally insignificant (and negative). The group-time average treatment effects range from 2.3% lower teen employment to 13.6% lower teen employment. The simple average (weighted only by group size) is 5.2% lower teen employment, and the average effect of a minimum wage increase across all groups that increased their minimum wage (corresponding to an estimate of θ_{sel}^O above) is 3.9% lower teen employment (see Panel (a) of Table 3). A two-way fixed effects model with a post treatment dummy variable also provides similar results, indicating 3.7% lower teen employment due to increasing the minimum wage. In light of the literature on the minimum wage these results are not surprising as they correspond to the types of regressions that tend to find that increasing the minimum wage decreases employment; see the discussion in Dube et al. (2010).

As in Meer and West (2016), there also appears to be a dynamic effect of increasing the minimum wage. For Illinois (the only state in the group that first raised its minimum wage in 2004), teen employment is estimated to be 3.4% lower on average in 2004 than it would have been if the minimum wage had not been increased. In 2005, teen employment is estimated to be 7.1% lower; in 2006, 12.5% lower; and in 2007, 13.6% lower. For states first treated in 2006, there is a small effect in 2006: 2.3% lower teen employment; however, it is larger in 2007: 7.1% lower teen employment.

Panel (a) of Table 3 reports aggregated treatment effect measures. First, we consider how the effect of increasing the minimum changes by the amount of time that the policy has been in place. These parameters paint largely the same picture as the group-time average treatment effects. The effect of